BIRDS OR BIRDERS?

Disentangling climate signals from survey effort in new bird distribution records across China

1,076
provincial new records
588 species · CBNR v1.1

175,640
species × province × year
cells in the risk set

649
first records modelled
392 species · 31 provinces (v2)

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A new record can mean the bird moved, or that someone finally looked

Every record needs a bird and a birder

Range maps are built from records.

A record = where birds are × where people look.

In the raw data, the two cannot be told apart.

In China, both grew fast after 2000

The climate warmed quickly.

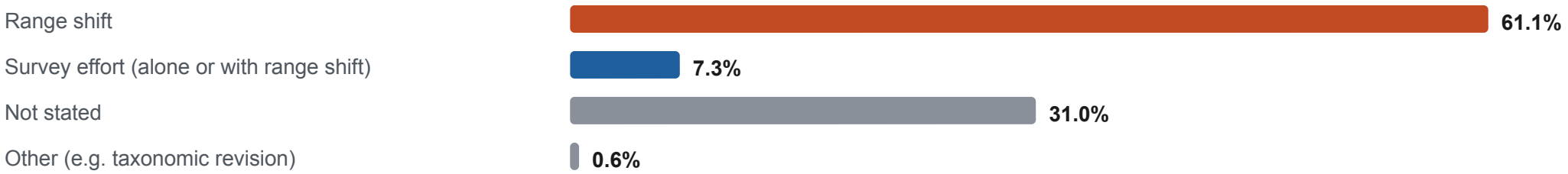
Birding visits rose about 200-fold (2002–2024).

A model with only one of them gives it the credit for both.

Our question

How much of the accumulation of new records reflects warming, and how much reflects survey effort?

How the original papers explain the CBNR v2 records (n = 1,029)

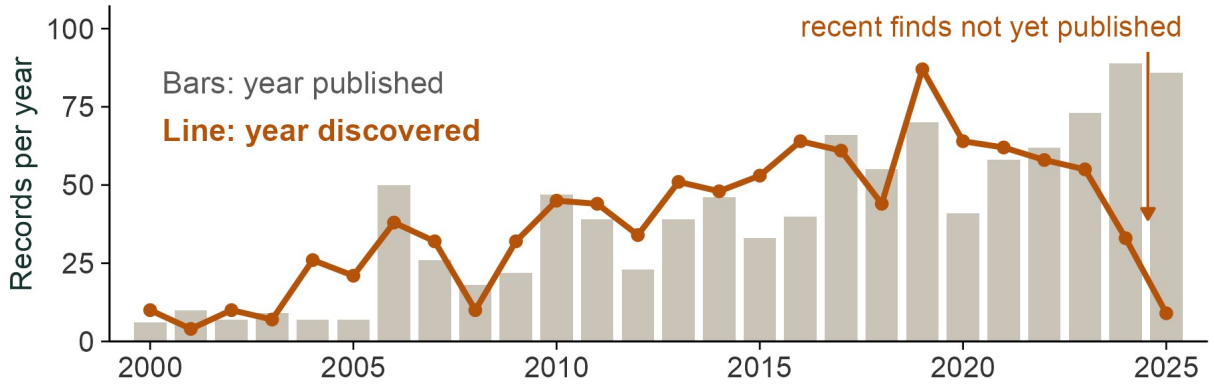


CBNR: provincial new bird records in China, 2000–2025

a 1,029 records across 33 provincial units



b Records reach print 2.1 years after discovery



CBNR, the dataset

v1.1: 1,076 records · 588 species

23 orders · 33 provincial units · 2000–2025

Mined from peer-reviewed papers, every record then checked by hand

What the model uses

The published v2 release (1,029 records)

Records dated to their discovery year

649 first records in 2002–2024

Open data · CC BY 4.0

doi.org/10.5281/zenodo.20809949

A discrete-time hazard model separates the two processes

For every species, in every province it could reach, in every year: is this the year of its first record?

first record ~ warming × effort + climate variability + offset(log completeness)
+ (1 | species) + (1 | province) + (1 | province × year)

binomial, cloglog link

Discrete-time hazard model: exp(β) is a hazard ratio (HR) per standard deviation

Complete risk set

Only provinces the species could reach
Leaves the risk set after its first record
Denominator = real opportunities

Species-referenced warming

Province anomaly − range anomaly
15-year moving window
1 SD = 0.18 °C

Survey effort

Birding visits per province and year
eBird and GBIF, 2.86 million records
1 SD = 7.9-fold more visits

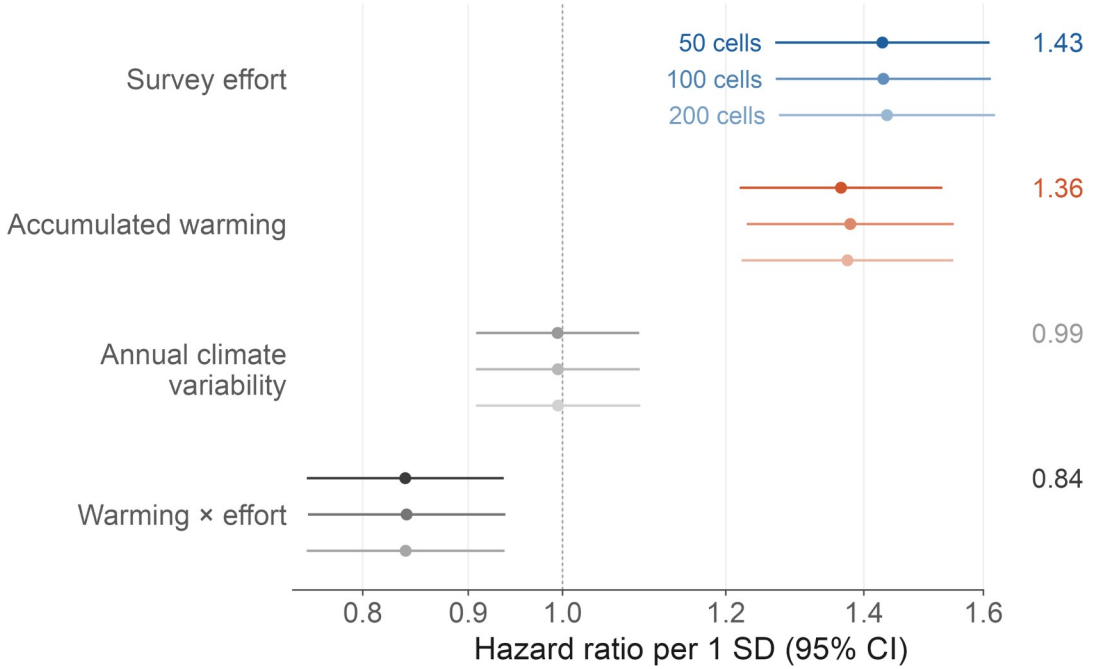
Province × year effect

The scale at which surveys happen
Bird races, campaigns, new reserves
The observation process, modelled

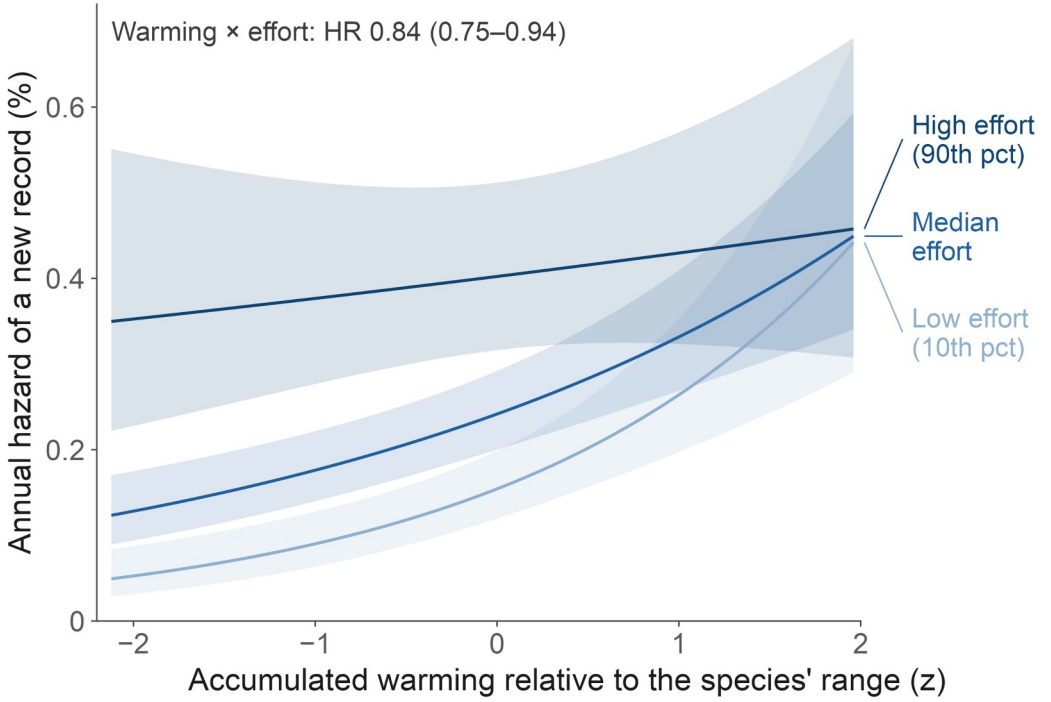
Records are dated to their discovery year; an offset corrects the most recent years for unpublished records.

Warming and survey effort contribute comparably and interact

Effort and warming contribute comparably



Warming matters most where effort is low



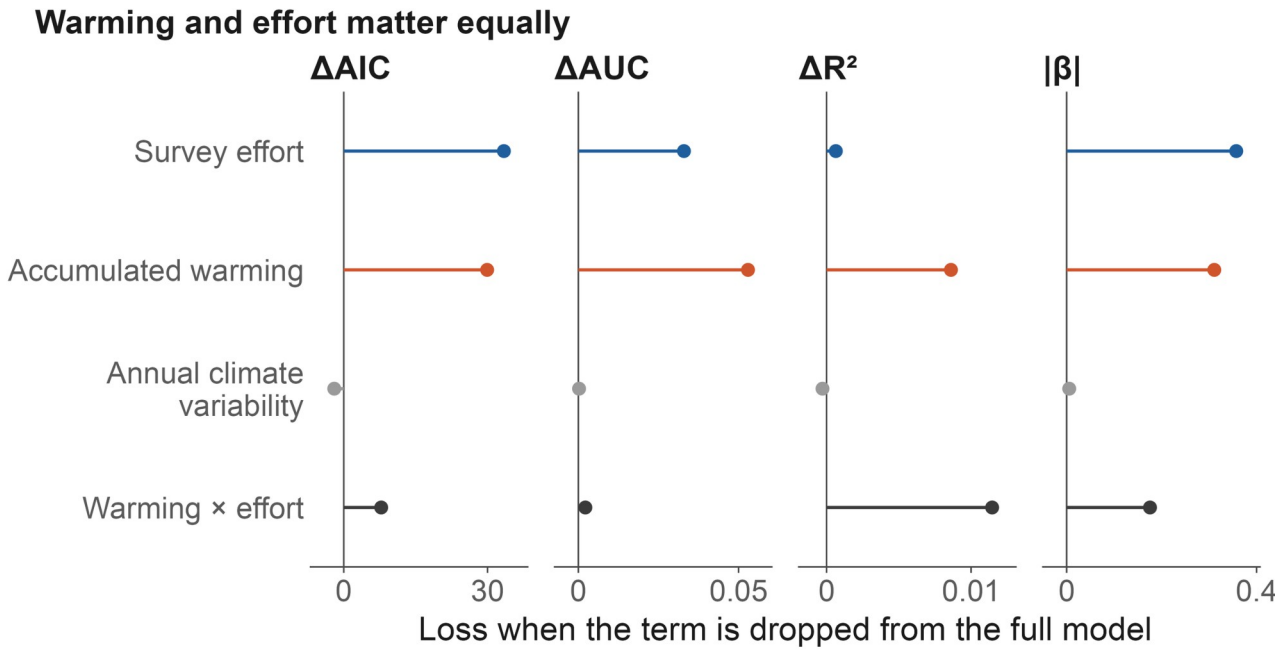
1.36
Accumulated warming
HR per SD (1.22–1.53)

1.43
Survey effort
HR per SD (1.27–1.61)

0.84
Warming × effort
strongest where few look

0.99
Annual variability
no effect

The balance holds across four criteria and twelve specifications



Cost of dropping each term

ΔAIC: warming 29.9 · effort 33.4

ΔAUC: warming 0.053 · effort 0.033

Conditional R^2 and |β| agree

Twelve specifications

SDM thresholds 50/100/200: stable

4 effort proxies × 3 missing-data rules

Effort HR 1.25–1.55 · warming 1.33–1.39

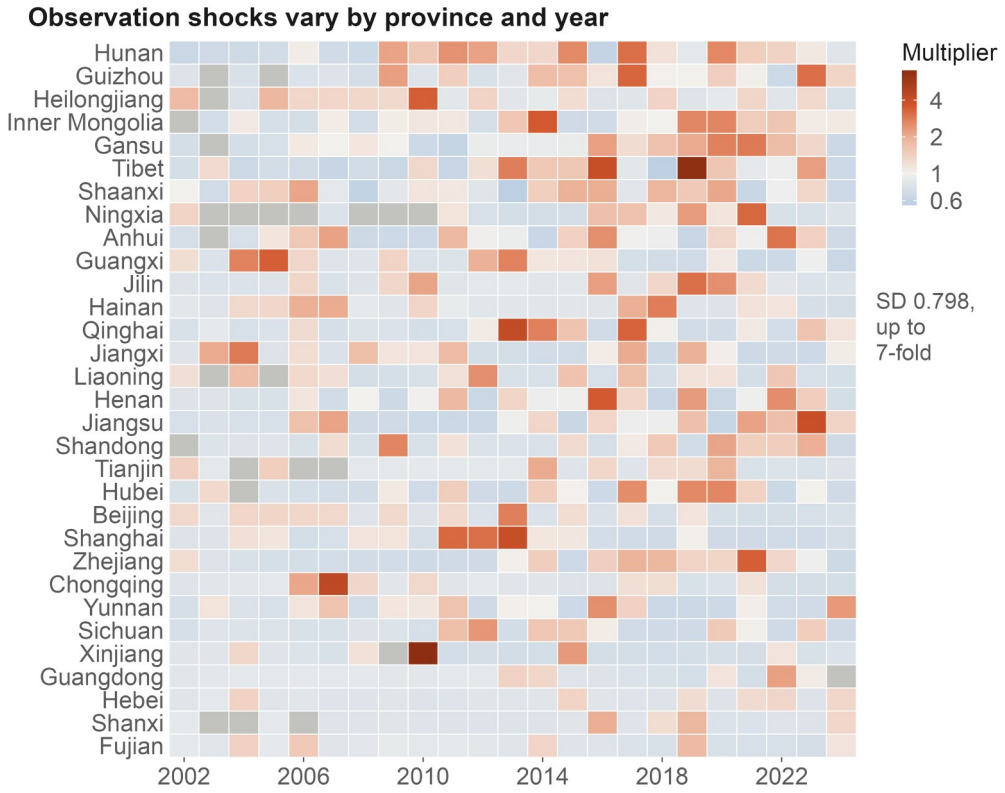
Design-preserving nulls

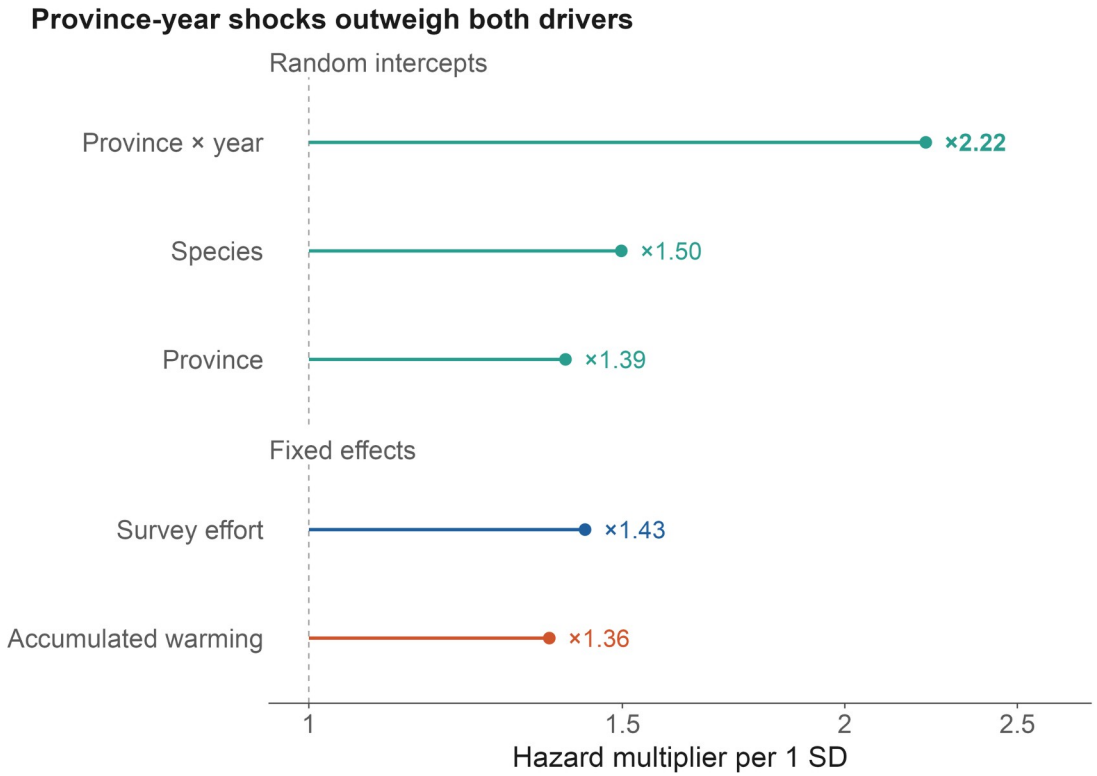
Warming null 1.16 → increment ×1.18

Shuffle effort timing: falls to 0.98

Province bootstrap: both lower bounds > 1

Pulses of survey activity are the largest source of variation





×2.22

Province × year

per SD: survey campaigns

×1.50

Species

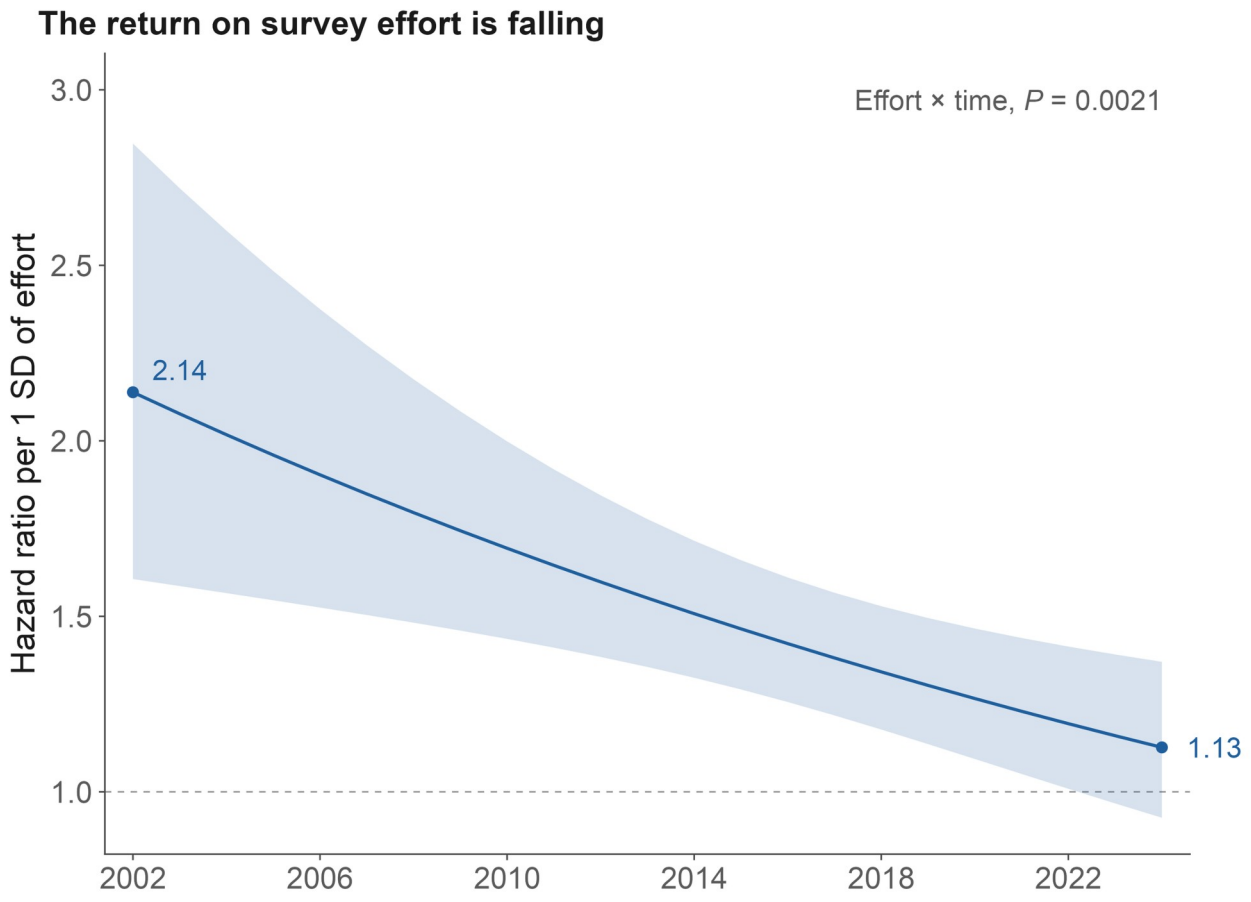
intrinsic detectability

×1.39

Province

regional baseline

The return on effort is falling; the effect of warming is not



Returns on effort are falling

2002: HR 2.14 (1.61–2.85)

2024: HR 1.13 (0.93–1.37)

Effort × time: P = 0.0021

Warming shows no such trend

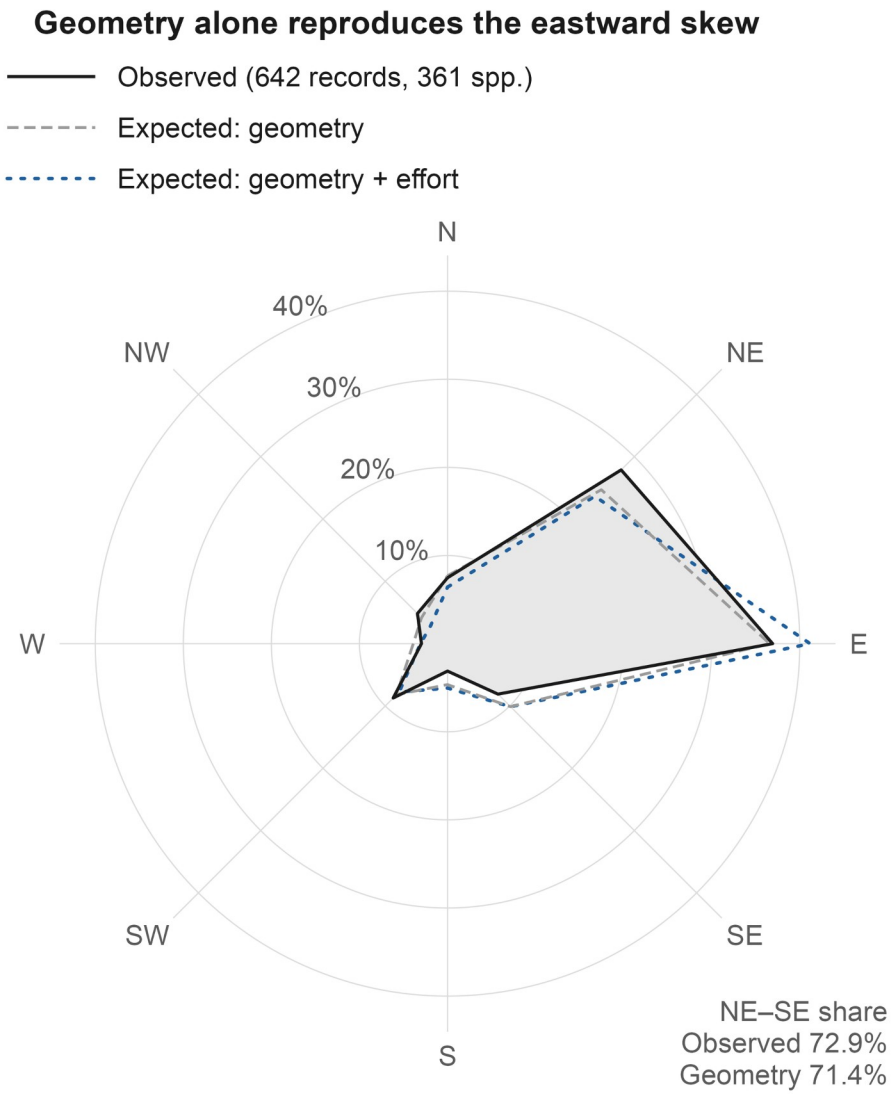
Warming × time: P = 1.00

Time-varying model: warming HR 1.37

A stock and a flow

Effort mines a finite stock of gaps;
warming keeps adding new candidates

The eastward bias is geometry; net of it, records lean slightly north



Raw bearings (642 records)

East 36.9% · north-east 27.9%

But China lies east of most ranges

Null 1 · geography only

East: 235 expected, 237 observed

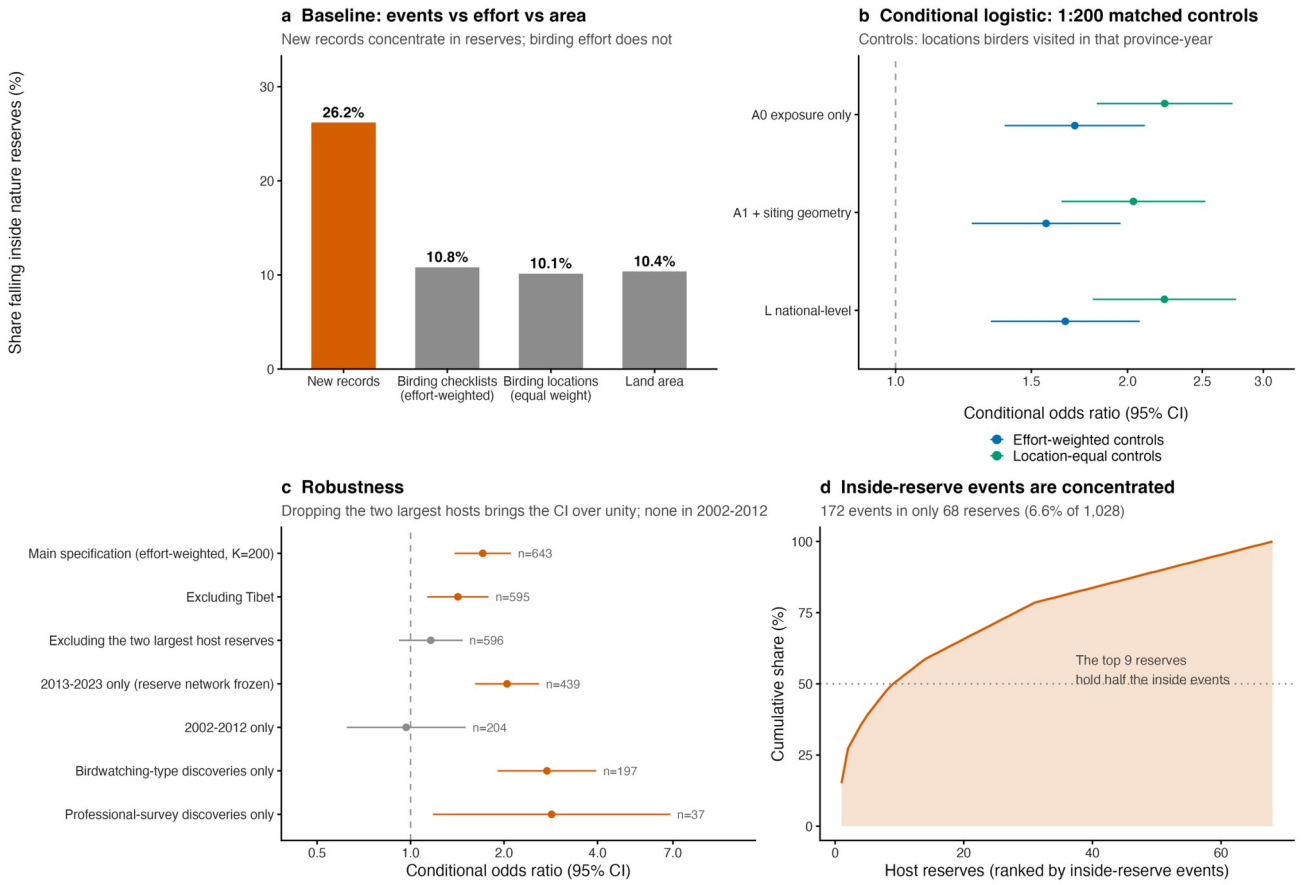
No sector differs

Null 2 · geography + birders

Northern sectors: 40% vs 33% expected

$\chi^2 = 26.8$, $P = 0.001$

Reserves look rich in new records because birders go there



An apparent 2.5-fold enrichment

Records inside reserves: 26.2%

Birding effort inside: 10.8%

Land area inside: 10.4%

Against where birders actually went

Same province and year: OR 1.57

Without the 2 largest reserves: 1.16

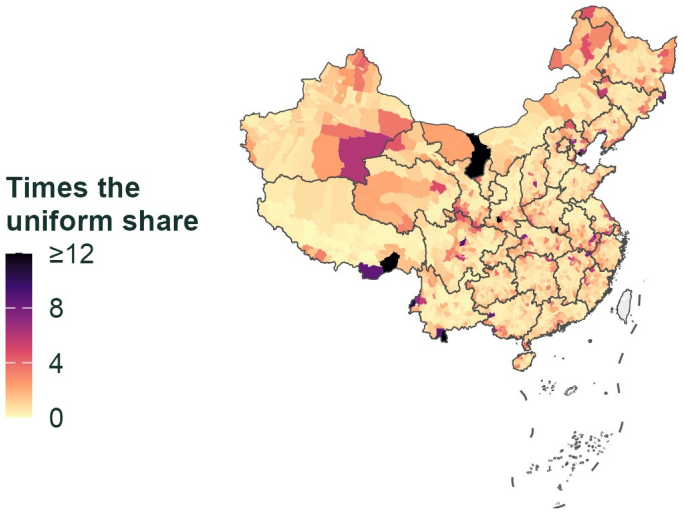
Before 2013: no effect (0.97)

More records ≠ better protection

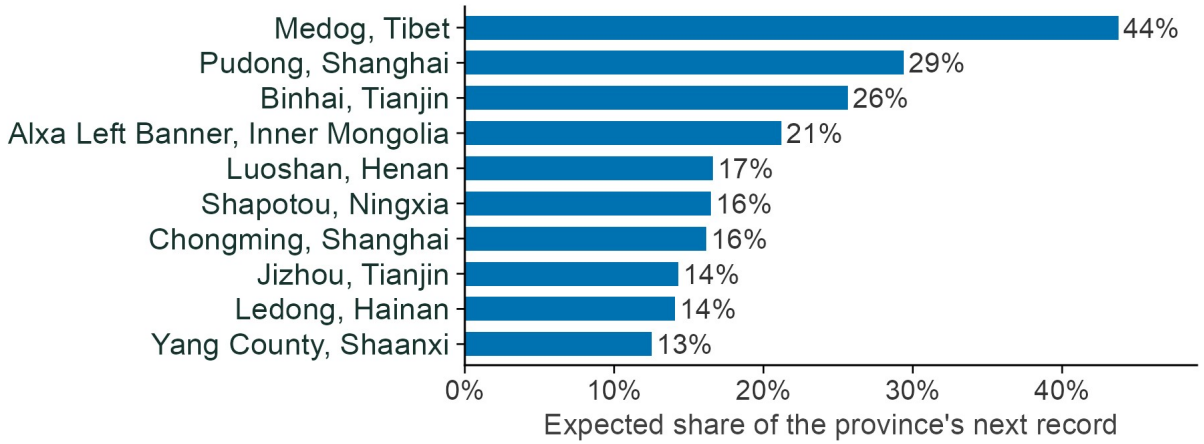
Record counts cannot measure how well a reserve works

Predictions can be pushed down to the county

a Relative chance of hosting the next provincial record



b Top 10 counties by expected share



Two steps

1 Will the province get a new record?

2 If so, which county?

Out-of-sample skill, 2019–2024

Top county right 16.1% (random ≈ 1%)

True county in the top 10%: 45.4%

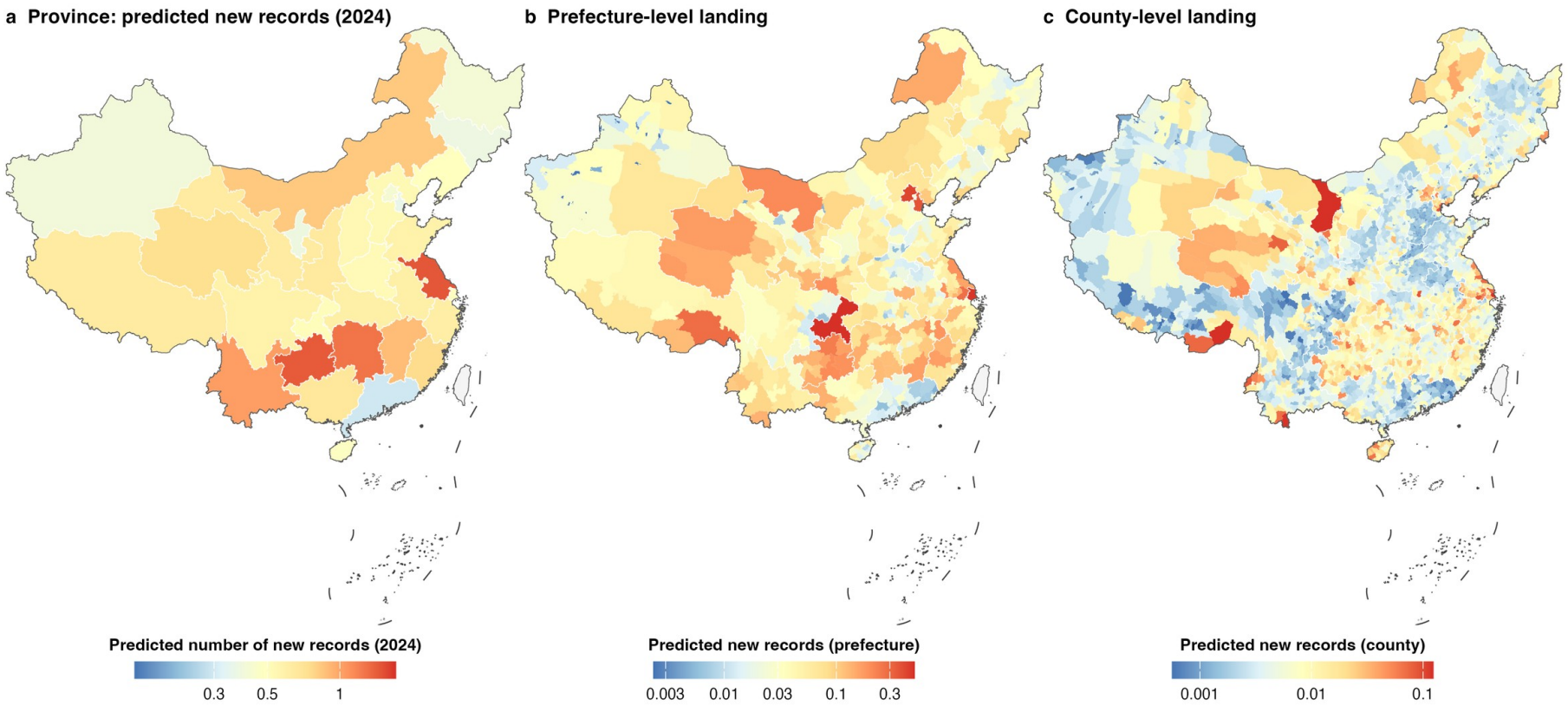
Leave one province out: 27 of 29

Birders will know these names

Medog · Pudong (Nanhui) · Chongming

Luoshan (Dongzhai) · Yang County

Province, prefecture and county maps from one consistent model



One model, three scales

Expected records (province) × share (unit)

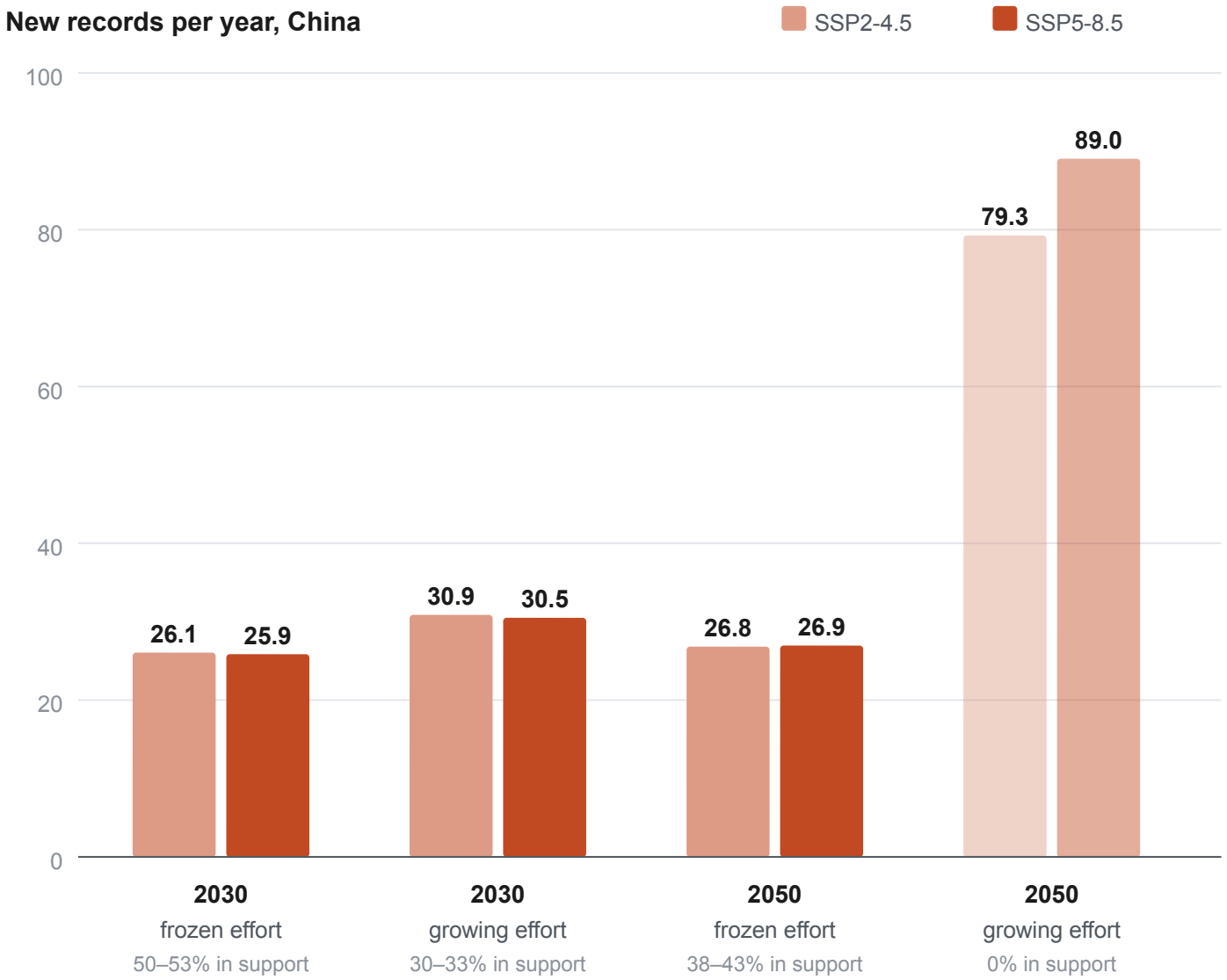
371 prefectures · 2,835 counties

Change the model family: maps barely move

Random forest / XGBoost: ρ = 0.59–0.89

Differences sit in the low-effort west

Future record counts follow survey effort, not the emission scenario



Two climate × two effort scenarios

Climate: SSP2-4.5 or SSP5-8.5

Effort: frozen, or +0.6 SD a decade

Horizons: 2030 and 2050

What changes the expected count?

Switch climate: barely anything

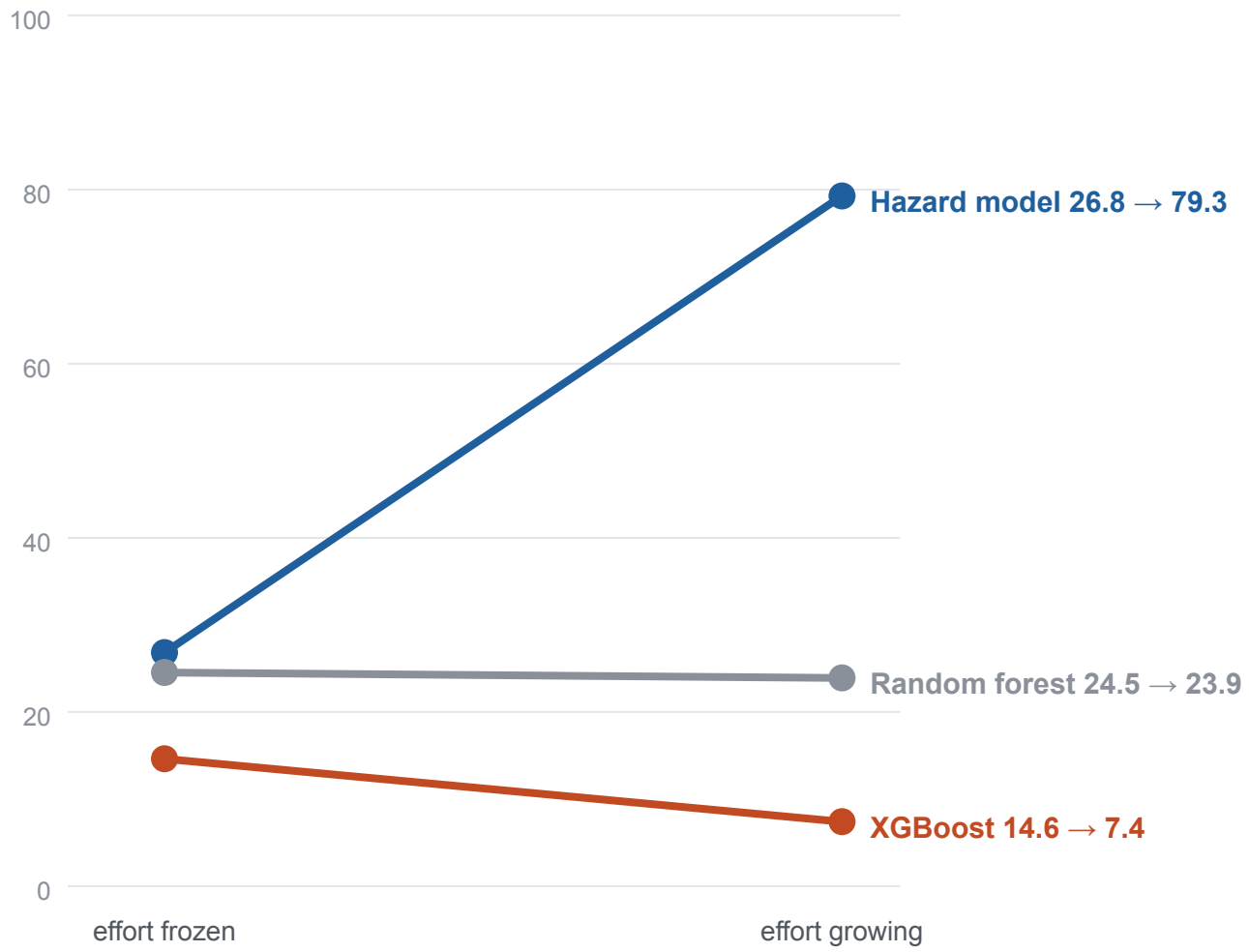
Keep effort growing: roughly triple

Most of that rise lies outside the data

Faded bars: no cell in support

Same futures, three model families: they diverge outside the data

New records per year in 2050 (SSP2-4.5)



Same scenarios, three model families

All agree on where records fall

ρ = 0.59–0.89 across units

They disagree on how many

Hazard model rises

Random forest stays flat

XGBoost falls

Not a contest between models

Outside the data nobody can check which direction is right

Data support, not ambition, sets the forecast horizon



Support shrinks fast (SSP2-4.5)

Cells inside the fitted range:

2030: 46% → 2050: 6% → 2080: none

Machine learning does not rescue it

Agree inside the range (ρ = 0.56)

Diverge outside it

Report where a model breaks

2030: usable

End of century: illustration, not forecast

Four limits that travel with the conclusions

Events are published records, not detections

We explain how records are documented, not range expansion itself.

Coordinates are precise to about 1 km at best

Reserve-overlap tests therefore use 0–5 km buffers as a sensitivity check.

Reserve polygons cover about a third of listed reserves

1,028 mapped reserves: no gap analysis, no 30×30 claims.

Projections leave the fitted support after 2030

Late-century maps are illustrations; the effort coefficient used is an upper bound.

Four take-home messages

1 New records carry a climate signal

Warming 1.36 vs effort 1.43 per SD

comparable, and separable in one model

2 Warming matters most where few look

Warming × effort = 0.84

under-surveyed regions hide the signal

3 Model the observation process

Province × year ×2.22 is the largest part;

returns on effort are falling

4 Futures depend on how we look

Emission scenario barely matters;

most projected change is extrapolation

Birds or birders? Both.

New records carry a climate signal,

but we can read it only once we model how we look.

Next

Reptiles and amphibians · monitoring priority = climate exposure × (1 − current coverage), mapped to counties

Put effort where warming is fast and records are few

Where the climate signal is strongest

Warming effect is largest at low effort

Low-effort provinces lie mostly in

the west and the interior

Where more effort adds little

Returns on effort fell from 2.14 to 1.13

Eastern hotspots are near saturation

Province level: IRR 0.71, P = 0.028

Monitoring priority

Send effort to climatically exposed, under-surveyed regions:

that is where new records carry the most climate information

Open data, open code, and thanks

Open data and code

CBNR (CC BY 4.0)

doi.org/10.5281/zenodo.20809949

github.com/dingchenchen6/

bird-newrecord-discovery-dated

Other data

Ranges: BirdLife International

Climate: CRU TS 4.09 · WorldClim 2.1 · CMIP6

Effort: 2.86 million GBIF / eBird records

Base map: approval no. GS(2019)1822

Acknowledgements

Birdwatchers across China, and every author whose records made CBNR possible.

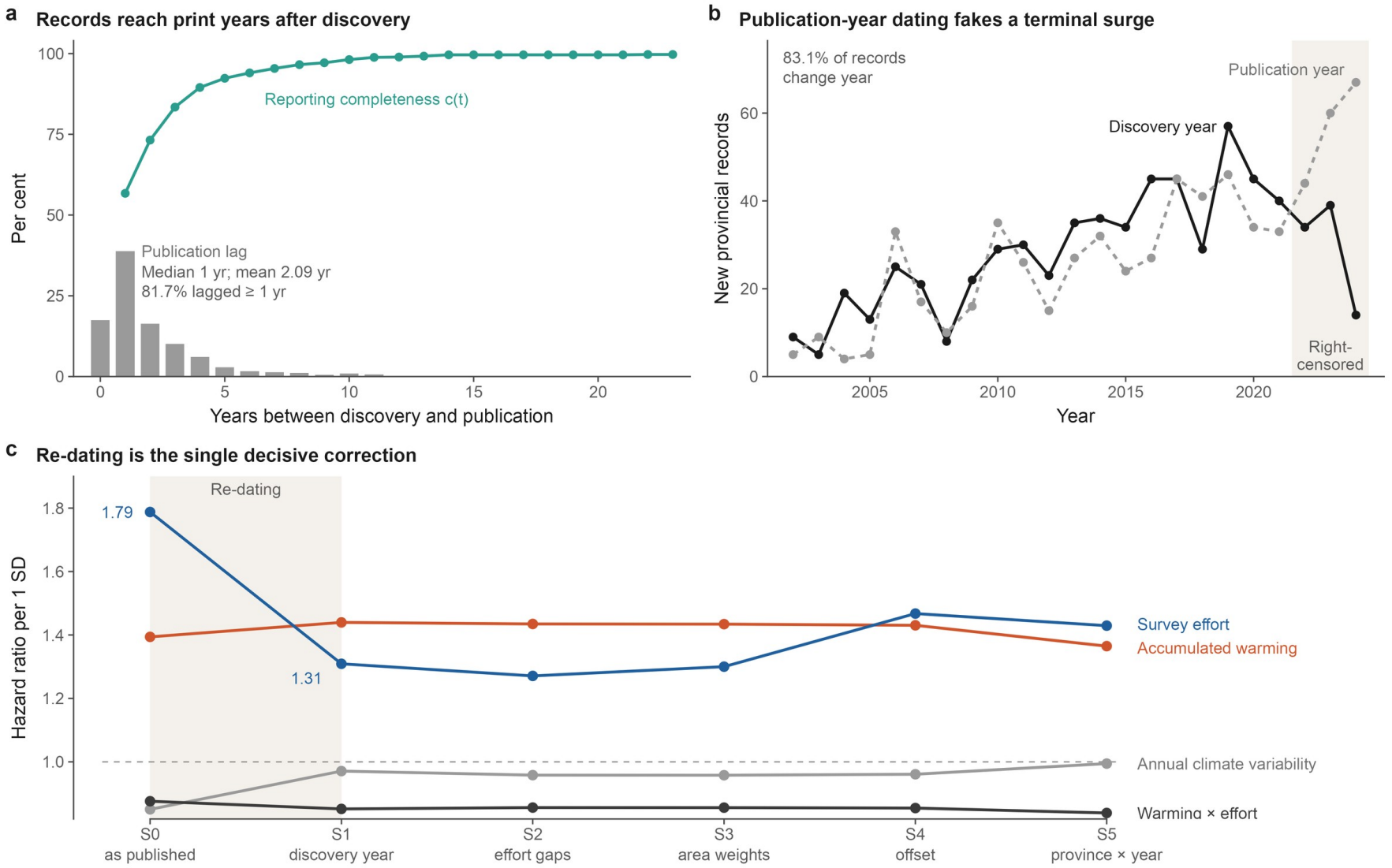
Thank you. Questions welcome

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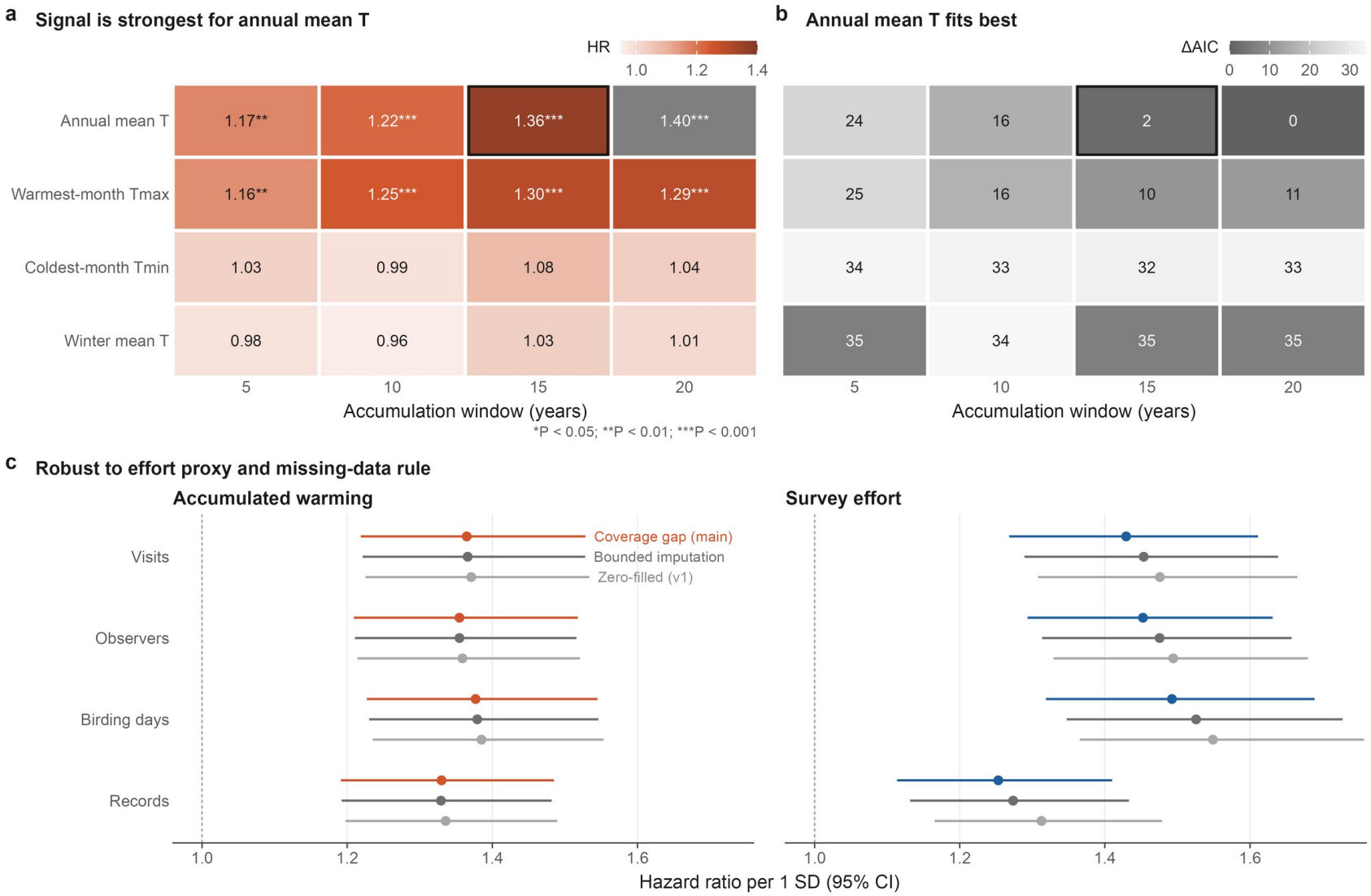
CBNR data

scan to download

Dating records to publication inflates effort and creates a phantom climate effect



Specification robustness: climate indicator, window and effort proxy



Model anatomy: climate referenced to each species' range

